

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger – Master NLP

Recap: Logistics of the ML4NLU-Module

- Don't forget that you must sign up for all exams!

There are two „exams“ for the module:

- The „written report“ is due end of February (soft deadline). The officially listed deadline is the end of March (end of semester) in PORTA. The hard deadline for the “written report” is end of March. If you miss the hard deadline, this automatically counts as a failed attempt, and you must sign up for a re-examination in the next semester.
- The written exam is a mixture of questions in the style of the “flipped classroom questions” and questions from the tutorial/exercise sheets (applied).

Sources

- Prompt Engineering Survey
 - <https://arxiv.org/pdf/2402.07927>
- Survey of Augmented Language Models
 - <https://arxiv.org/pdf/2302.07842.pdf> (Section 3)
- Deep Dive In-context Learning
 - <https://youtu.be/70OCV8XfMbo?si=jvLw4ty3Y7EFdTKa>
 - <https://youtu.be/0mXbM2j3Dzs?si=PUJUZeJHNnIxIYb7>
- Deep Dive RAGs
 - <https://www.youtube.com/watch?v=PrYuqtT43BE>

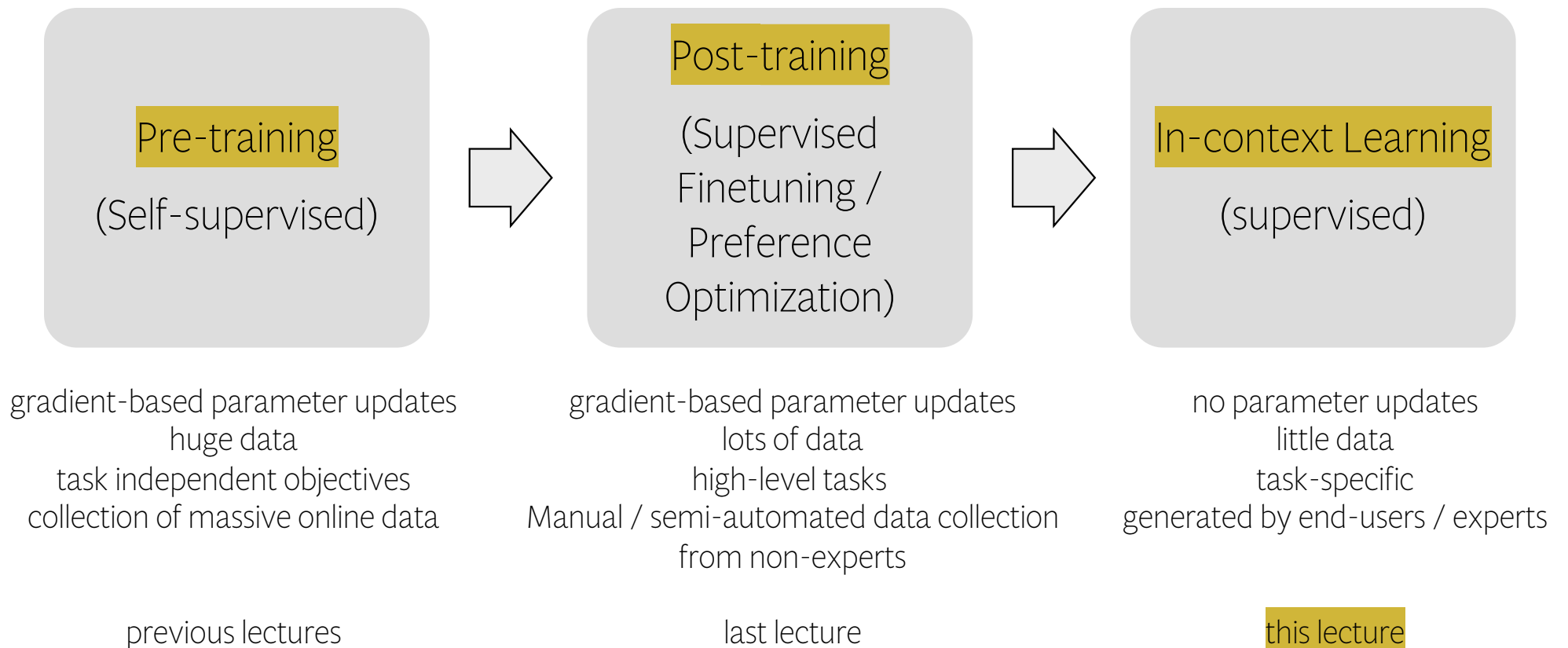
Flipped Classroom Questions

- What is the difference between “few-shot supervised learning” and “few-shot in-context learning”?
- Name typical elements of a prompt?
- Describe a few popular prompting techniques
- Name tasks that plain LLMs don’t handle well, but other tools do
- Name techniques that mitigate those shortcomings of LLMs

Prompt Engineering + Tool Augmentation

Lecture 10

Recap: Training Steps of LLMs



Terminology: In-Context Learning

<https://youtu.be/7OOCV8XfMbo?si=r-HwHJ1h6HI2muKr&t=167>

- **In-context learning:** A frozen LM performs a task only by conditioning on the prompt text.
- **Few-shot in-context learning:** (1) The prompt includes examples of the intended behavior, and (2) no examples of the intended behavior were seen in training.
 - ▶ We are unlikely to be able to verify (2).
 - ▶ “Few-shot” is also used in supervised learning with the sense of “training on few examples”. The above is different.
- **Zero-shot in-context learning:** (1) The prompt includes no examples of the intended behavior (but it can contain other instructions), and (2) no examples of the intended behavior were seen in training.
 - ▶ We are unlikely to be able to verify (2).
 - ▶ Formatting and other instructions seem like a gray area, but we will allow them in this category.

Terminology: Prompting

- Prompt engineering is the art of designing queries that effectively elicit desired responses from LLMs.
- Prompting is **inference** NOT training (no parameters are updated)
- Prompts can include examples/demonstrations of the desired behavior. This is called “few-shot prompting” or “in-context learning”. This still is not model training (in the sense of updating the model’s parameters). If parameters are updated, it is “post training / instruction tuning” (see last lecture).
- Prompts without examples might be called “zero-shot in-context learning”.



Elements of prompting

Section 10.1.1

Recap: Elements of Instruction Tuning

```
{  
  "instruction": "Write a limerick about a  
                 pelican.",  
  "input": "",  
  "output": "There once was a pelican so fine,  
             \nHis beak was as colorful as  
sunshine,\nHe would fish all day,\nIn  
a very unique way,\nThis pelican was  
truly divine!\n\n\n",  
},
```

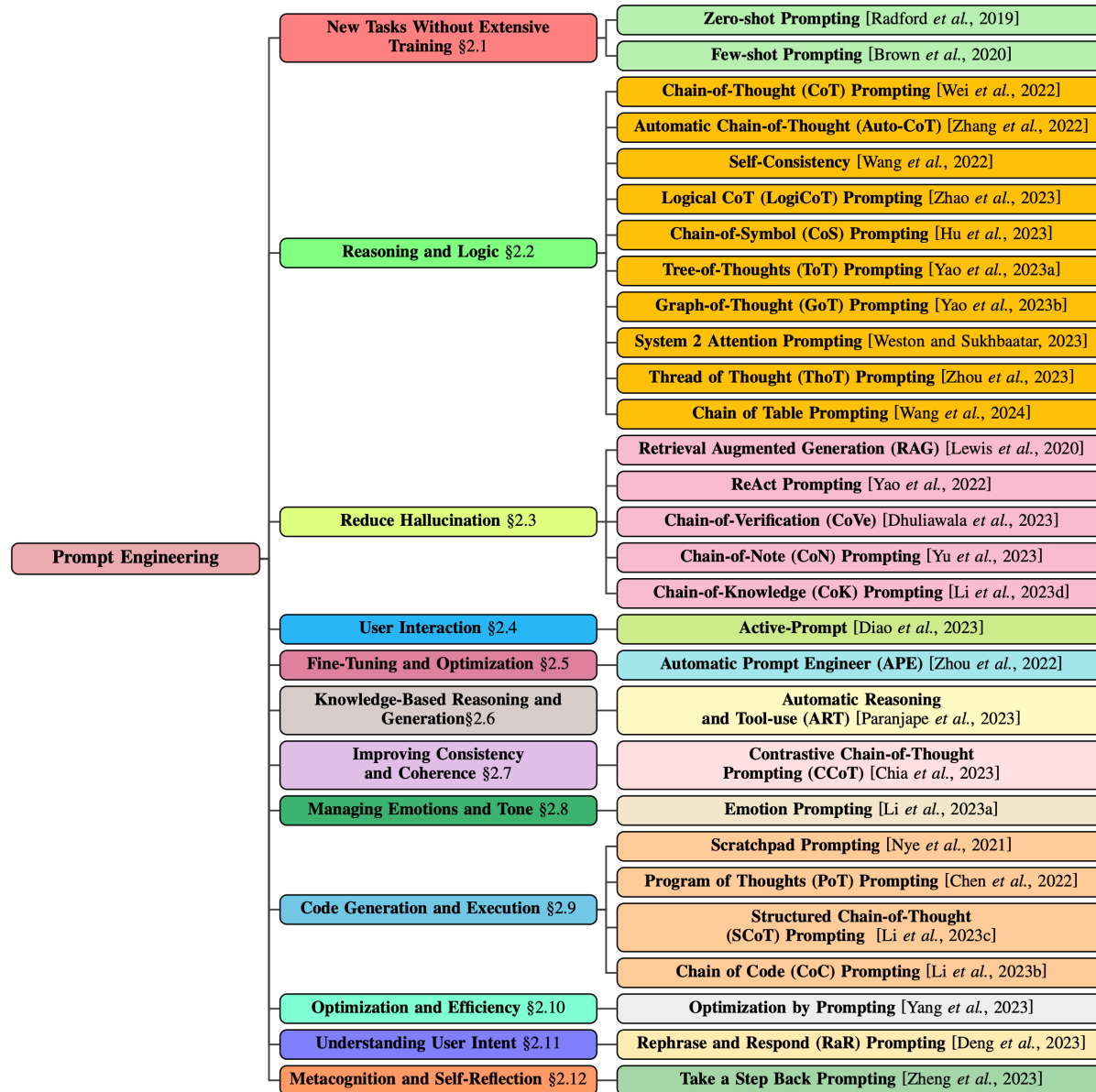
```
{  
  "instruction": "Identify the odd one out from  
                 the group.",  
  "input": "Carrot, Apple, Banana, Grape",  
  "output": "Carrot\n\n",  
},
```

Typical elements of a prompt

1. Instruction or Question
 - Either or needs to be present
2. Context
 - Context can be any kind of auxiliary information
3. Examples/Demonstrations
 - => in-context learning
4. Formating guidelines
 - Example:
 - “Q: ...”
 - “A: ...”

Prompting techniques

Section 10.1.2



<https://arxiv.org/pdf/2402.07927>

Demonstrations <https://youtu.be/0mXbM2j3Dzs?si=eLGEbKX1AXNn9NjF&t=47>

Context: Kermit is one of the stars of Sesame Street.

Train/Retrieve

Q: Who is Kermit?

Train

A: Kermit is the one of the stars of Sesame Street.

Train/Retrieve/Generate

Context: Bert is a Muppet who lives with Ernie.

Retrieve

Q: Who is Bert?

Given

A:

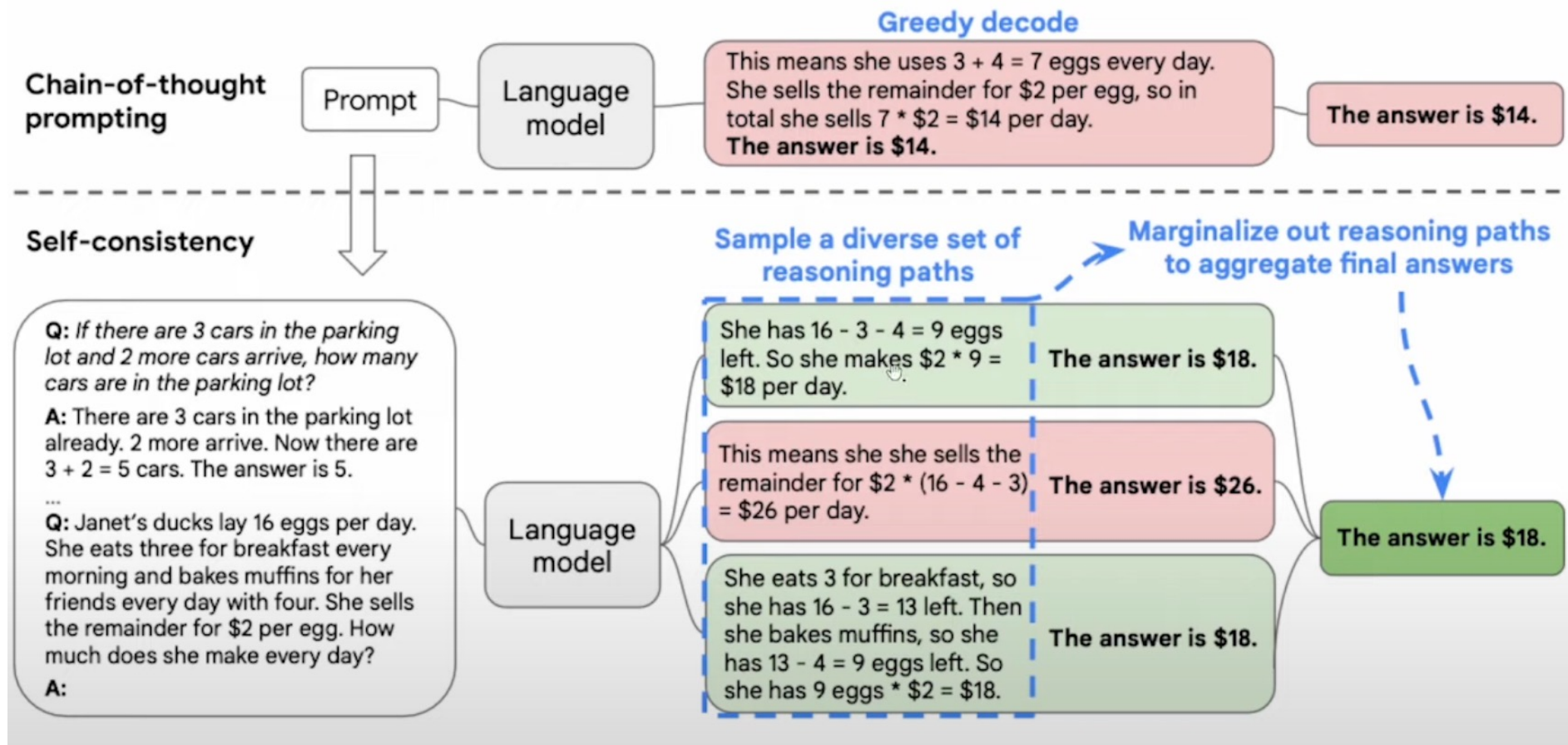
Chain of thought <https://youtu.be/0mXbM2j3Dzs?si=YzVpZUMiObfXDipE&t=551>

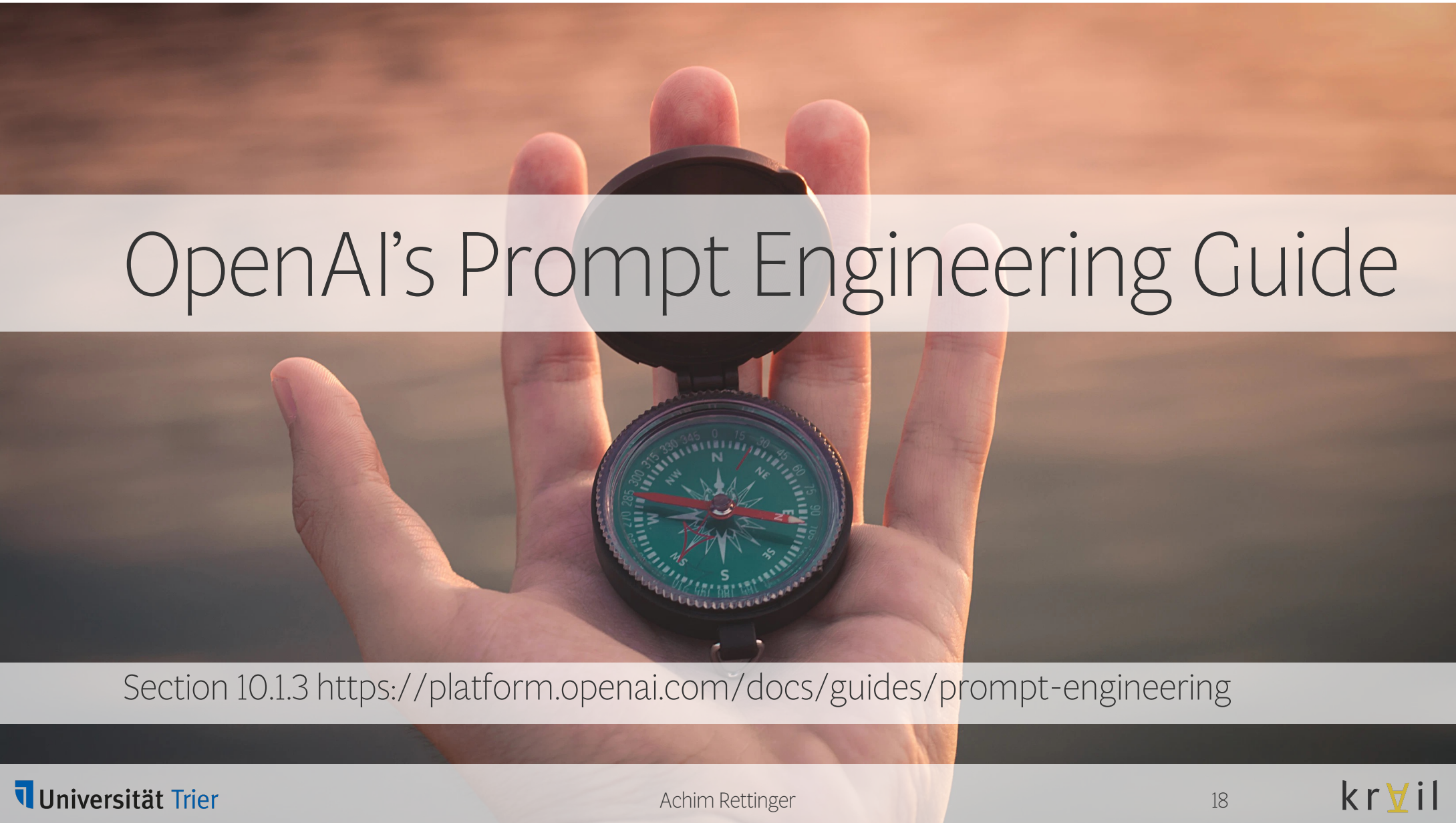
- Naïve approach:
 - Q: A: Let's think step by step. Therefore, the answer is..."

- Better:
 - Use demonstrations

Standard Prompting	Chain-of-Thought Prompting
Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?	Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?
Model Output A: The answer is 27. ❌	Model Output A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Self consistency <https://youtu.be/0mXbM2j3Dzs?si=Xkt7pQWLgKOhfh65&t=721>

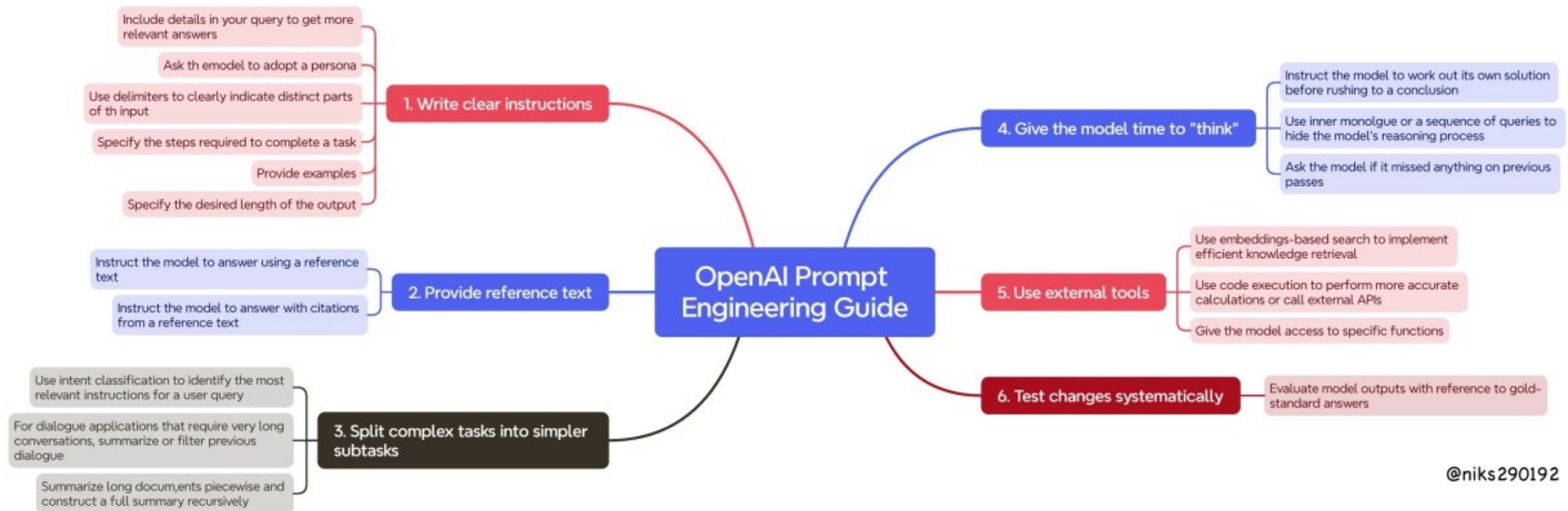


A hand holding a compass against a sunset background. The compass is open, showing a green face with white markings and red hands. The background is a warm, orange-hued sunset over a body of water.

OpenAI's Prompt Engineering Guide

Section 10.1.3 <https://platform.openai.com/docs/guides/prompt-engineering>

OpenAI's recommendations for getting better results



@niks290192

Selected Tactics I

- Ask the model to adopt a persona
- Use delimiters to clearly indicate distinct parts of the input
- Provide examples
- Provide reference text
- Use system message to specify the persona used by the model in its replies.
- >Abstract: ""insert abstract here"">
- See “demonstrations”
- >Use the provided articles to answer questions. If the answer is not in the articles, write "I could not find an answer."<

Selected Tactics II

- Use intent classification to identify the most relevant instructions for a user query
- Give the model time to "think"
- Use external tools
- Evaluate model outputs with reference to gold-standard answers
- First classify the type of query and use that classification to determine which instructions are needed.
- See "chain-of-thought"
- See next section
- Automatically quantify the performance of the model by letting it count how often it was correct



Limited Capabilities of LLMs

Section 10.2

Limits of LLMs

1. LLMs are bad/inefficient at stuff that other tools are perfect/efficient, like a calculator
2. LLMs are bad at (up-to-date / out-of-parametric-memory) information sources

LLMs can use tools (<https://arxiv.org/pdf/2302.04761.pdf>)

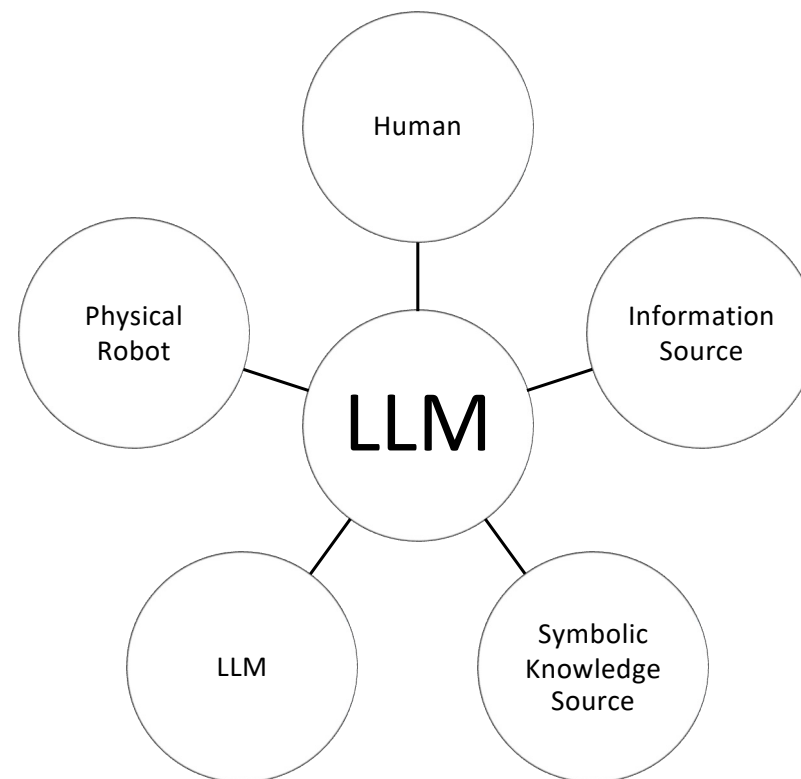
Out of 1400 participants, 400 (or **[Calculator(400 / 1400) → 0.29]** 29%) passed the test.

The name derives from “la tortuga”, the Spanish word for **[MT(“tortuga”) → turtle]** turtle.

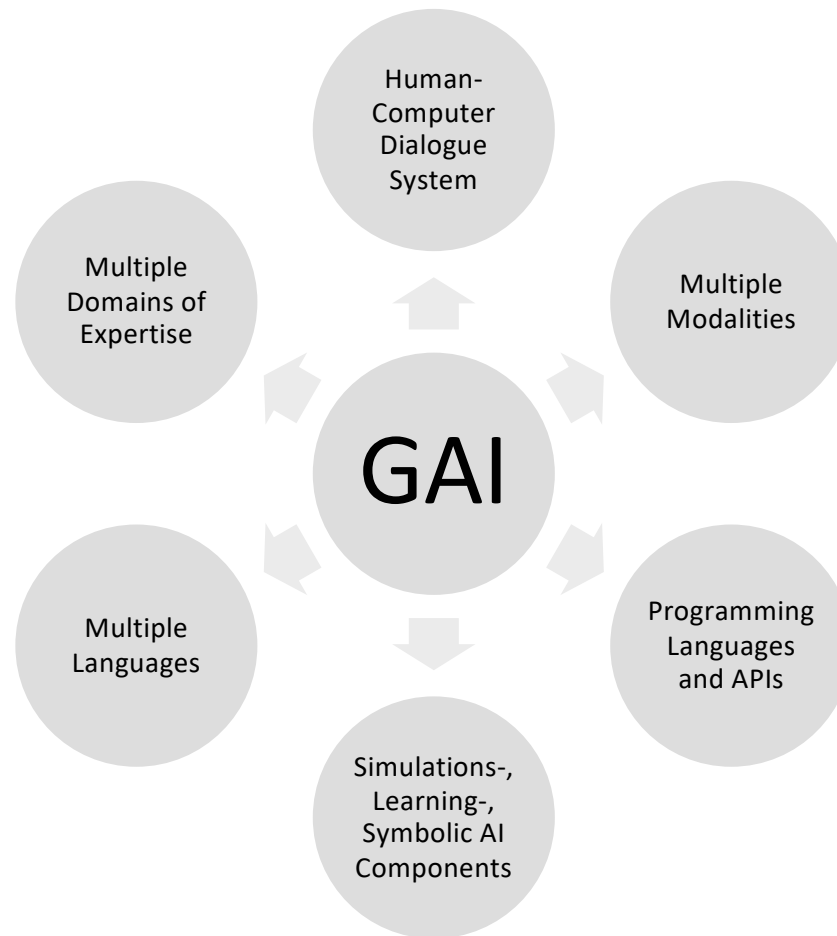
The Brown Act is California’s law **[WikiSearch(“Brown Act”) → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.]** that requires legislative bodies, like city councils, to hold their meetings open to the public.

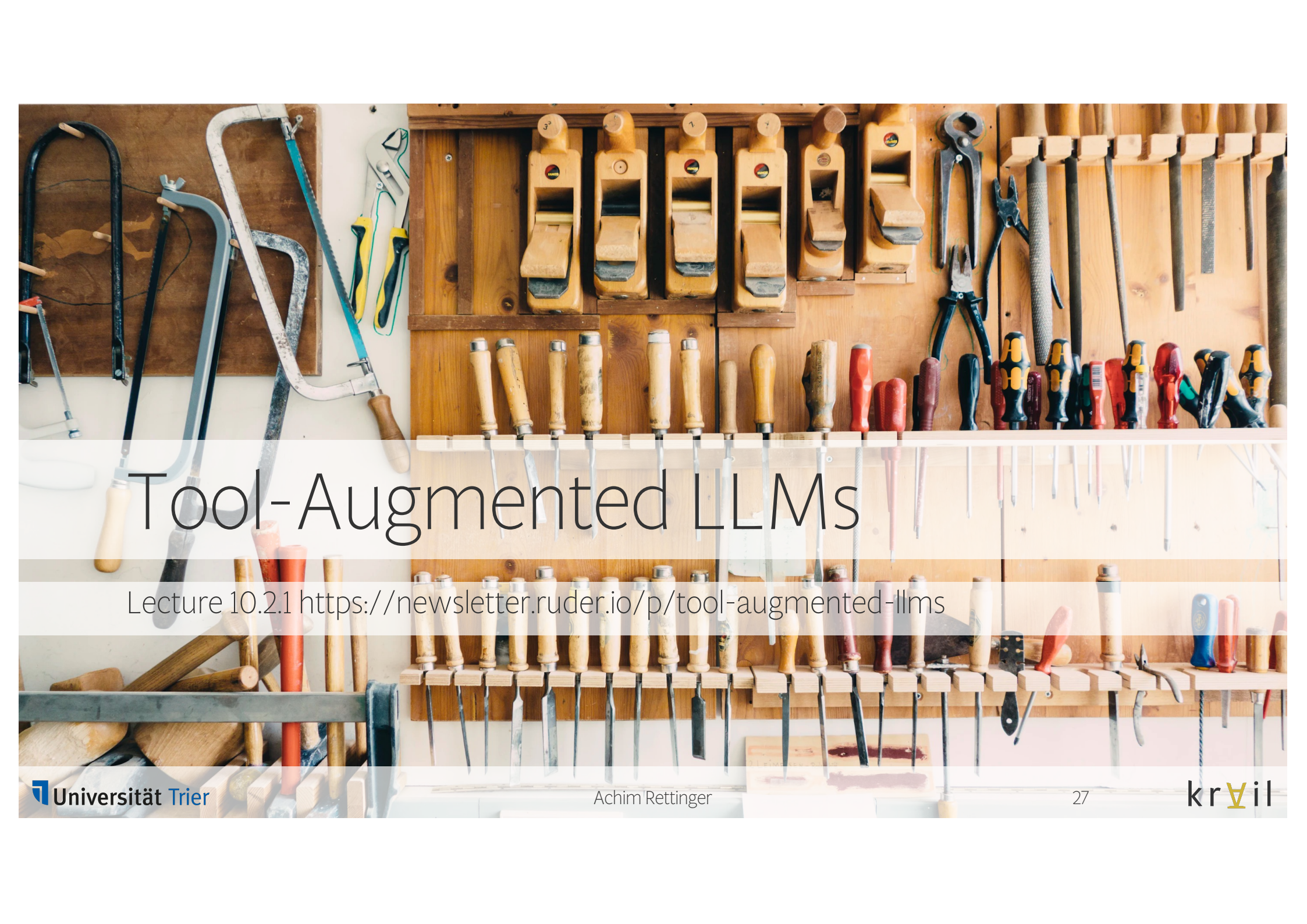
LLMs are translator by nature

Their underlying neural network architecture is “Transformers”



LLMs can be “the glue” of hybrid systems





Tool-Augmented LLMs

Lecture 10.2.1 <https://newsletter.ruder.io/p/tool-augmented-llms>

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How to teach tool use – Option 1 and 2

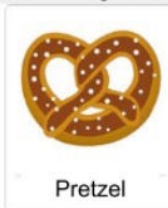
Demonstration

Description: examples of questions and the tool-use plan.

- **Question:**
Which property do these objects have in common?



Chips



Pretzel



Fries

Tool-use Plan:

Text Detector → Knowledge Retriever → Solution Generator

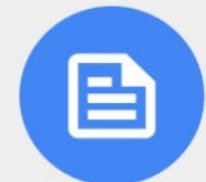
- **Question:** [...]
- **Tool-use Plan:** [...]
- **Question:** [...]
- **Tool-use Plan:** [...]



Documentation

Description: available tools and their functionalities.

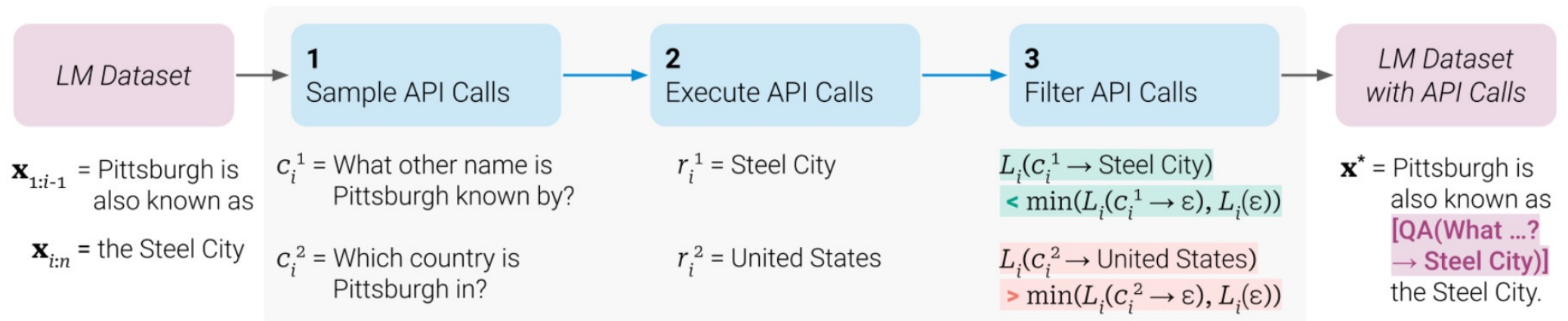
- **Text Detector:**
It detects the text in an image [...]
- **Knowledge Retriever:**
It retrieves relevant knowledge [...]
- **Search Engine:**
It searches the web for relevant info [...]
- **Image Captioner:**
It generates a caption for an image [...]
- ...



<https://arxiv.org/pdf/2308.00675.pdf>

How to teach tool use – Option 3

Fine-tuning on data augmented with API calls

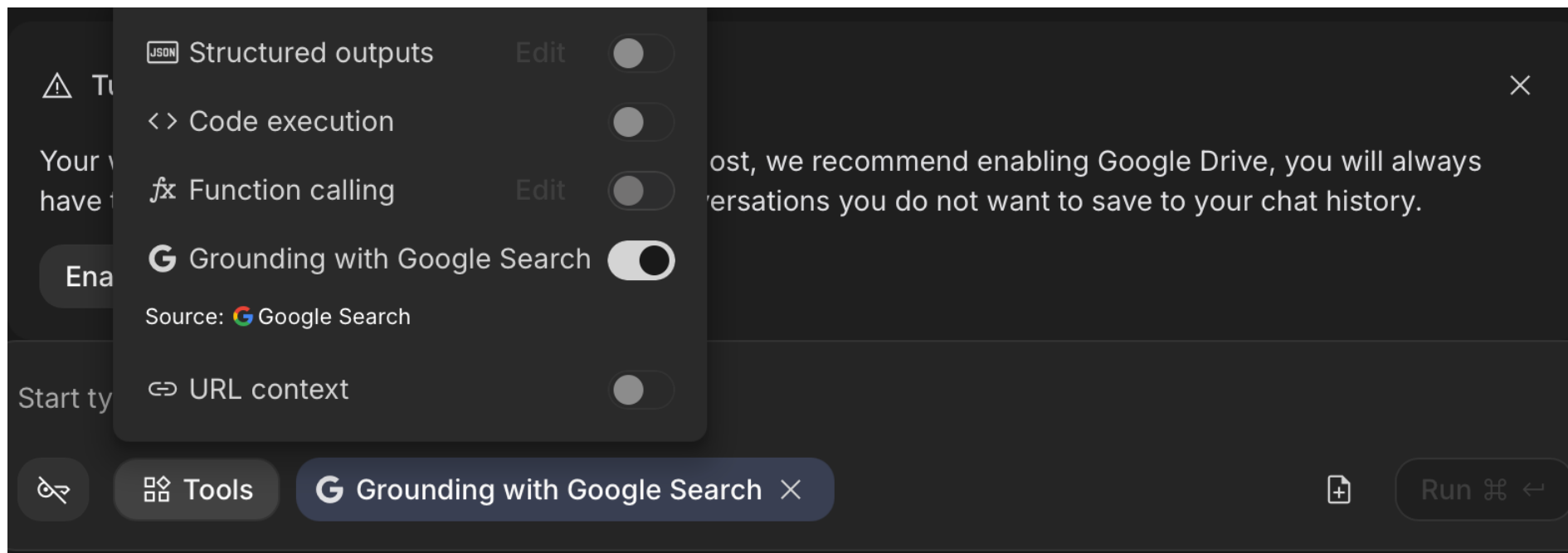


<https://arxiv.org/pdf/2302.04761.pdf>

Limits of LLMs

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Accessing news via web search





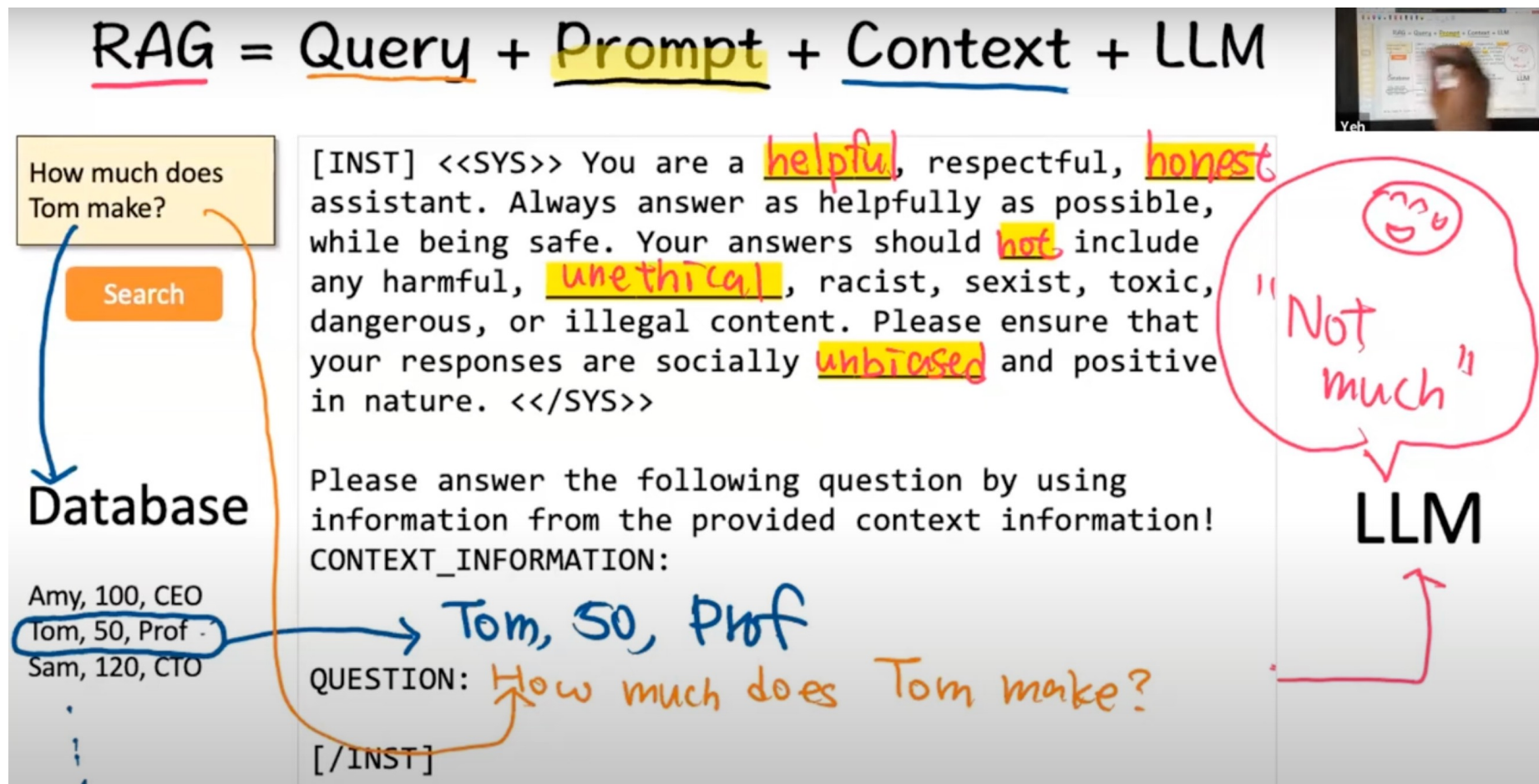
Retrieval Augmented Generation

Section 10.2.2

Common Requirements in Commercial Services

- Information from company internal sources need to be included in the generated output
- Internal sources should be hosted internally and not be part of the parametric memory of the LLM
- Information sources are dynamic and need to be updated constantly, e.g. news articles
- The sources of information need to be made explicit, e.g., by references

Basic Idea of RAG <https://youtu.be/PrYuqtT43BE?feature=shared>



How is the database entry selected? -> Vector data base

<https://youtu.be/RbrTFyvhIwQ?si=yypbH00ITkCs70piX>

Which record is relevant to the query “cat”?

Query

cat

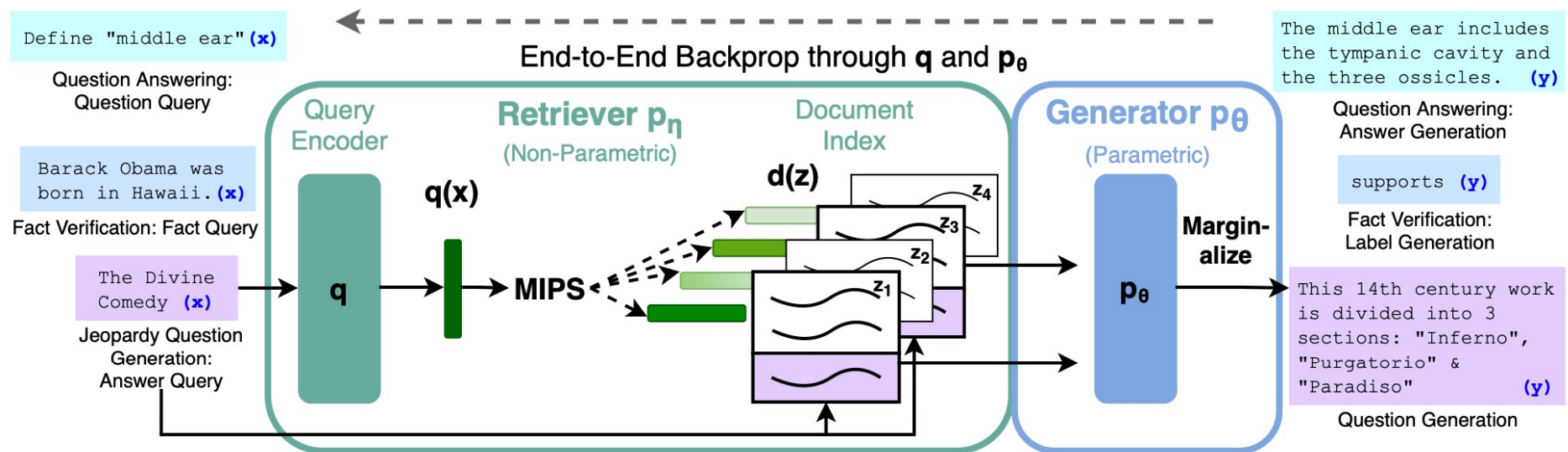
1	2	0
---	---	---

id	name	size	pop	emb			
1	dog	2	900	<table><tr><td>2</td><td>1</td><td>0</td></tr></table>	2	1	0
2	1	0					
2	bat	1	10000	<table><tr><td>0</td><td>1</td><td>2</td></tr></table>	0	1	2
0	1	2					

Recap: Dense Vector-based DSM

- Use an algebraic vector representation
 - Continuous values
 - Dense, fixed length, low-dimensional (100 – 1000)
 - Latent / hidden
- aka “embedding”
- *Adjust* the numbers so that each vector is good in predicting words in the context:
- Apple $<[1 \ 1]^T, \text{ Orange } [1 \ 0.9]> = 1.9$

Retrieval Augmented Generation (<https://arxiv.org/pdf/2005.11401.pdf>)



“The future of language is about the development of language models...”¹

1

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